

Math 521, Section 2
Homework 10
Selected Solutions

Problem 3. Let (X, d) be a metric space.

- (a) Prove that if $A \subseteq B \subseteq X$, A is closed in X , and B is compact, then A is compact.
- (b) Prove that if $C, D \subseteq X$ are both compact, then $C \cap D$ is compact.

Solution: (a) Let $\{U_i\}_{i \in I}$ be an open cover of A . As A is closed, then A^c is open. So $\{A^c\} \cup \{U_i\}_{i \in I}$ is an open cover for B , since each set is open and we have

$$B \subseteq X = A^c \cup A \subseteq A^c \cup \bigcup_{i \in I} U_i.$$

As B is compact, there must exist a finite subcover for B . We may write this subcover as $\{A^c\} \cup \{U_i\}_{i \in J}$ for some finite set $J \subseteq I$, because if A^c is not part of the subcover, then we can add A^c and still have a finite subcover. Altogether, this yields

$$B \subseteq A^c \cup \bigcup_{i \in J} U_i.$$

As A is a subset of B and is disjoint from A^c , then we must have

$$A \subseteq \bigcup_{i \in J} U_i.$$

So $\{U_i\}_{i \in J}$ is a finite subcover of A .

(b) As C and D are compact, then they are closed by Lecture 27. Therefore $C \cap D$ is closed as well. Altogether, we see that $C \cap D$ is a closed subset of a compact set C , and thus $C \cap D$ is compact by part (a). □

Problem 4. Let (X, d) be a metric space and let $Y \subseteq X$. We call the set Y complete if the subspace Y with the induced metric is complete. Prove that if Y is compact, then Y is complete.

Solution: Let $\{y_n\}_{n \geq 1}$ be a Cauchy sequence in Y . We want to show that $\{y_n\}_{n \geq 1}$ converges to some point in Y .

As Y is compact, then by Lecture 26 there exists a subsequence $\{y_{k_n}\}_{n \geq 1}$ which converges to some point $y \in Y$.

To finish, we will show that the whole sequence $\{y_n\}_{n \geq 1}$ converges to y . Fix $r > 0$. As y_{k_n} converges to y , there exists N_1 so that

$$d(y_{k_n}, y) < \frac{r}{2} \quad \text{for all } n \geq N_1.$$

As $\{y_n\}_{n \geq 1}$ is a Cauchy sequence, then there exists N_2 so that

$$d(y_n, y_m) < \frac{r}{2} \quad \text{for all } n, m \geq N_2.$$

Set $N = \max\{N_1, N_2\}$. Then, for $n \geq N$ we have $k_n \geq n \geq N$, and so

$$d(y_n, y) \leq d(y_n, y_{k_n}) + d(y_{k_n}, y) < \frac{r}{2} + \frac{r}{2} = r.$$

As $r > 0$ was arbitrary, we conclude that $\{y_n\}_{n \geq 1}$ converges to y . □